

Forecasting Criminal Arrests

A Time Series Analysis of New York City Arrest Data 2012-2017

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Abstract

This paper explores the application of time series analysis to forecast daily criminal arrests in New York City, aiming to assist law enforcement in efficient resource allocation. Using publicly available NYPD arrest data aggregated from 2012 to 2017, we modeled daily arrest counts as a stochastic process exhibiting strong weekly seasonality. We evaluated four variations of the Seasonal Autoregressive Integrated Moving Average (SARIMAX) model, testing the effects of log-transformation and the inclusion of exogenous variables representing holidays and significant events.

Our results indicate that incorporating exogenous variables consistently improves model performance by accounting for external disruptions. While the log-transformed model with exogenous variables (LOG-EXO) achieved the best statistical metrics (MAE 0.10, R2 0.6554), visual analysis of the residuals revealed high autocorrelation, suggesting a poor structural fit. Conversely, the model using the original scale with exogenous variables (OG-EXO) demonstrated lower residual autocorrelation, indicating a more accurate representation of the underlying data patterns despite slightly higher numerical errors. Consequently, the OG-EXO model was selected for the final analysis.

that there is sufficient coverage to handle criminal activity. Accurate forecasting of crime rates can aid in this endeavor, allowing law enforcement agencies understand patterns and trends in criminal activity, and allocate resources accordingly.

To this end, many police departments around the world collect and publish data on arrests made within their jurisdictions. The New York Police Department (NYPD) is no exception; it maintains a publicly available list of arrests [3] made in the city starting from the year 2006. This dataset is updated annually, and contains the particulars of all the aforementioned arrests, including the date of arrest. We have modified this dataset to obtain a more useful time series, by aggregating the number of arrests made on each day from January 1, 2012 to December 31, 2017. The resulting time series contains a total of 2192 data points, with each value corresponding to the number of arrests made on a particular day in New York City.

We will be using this dataset to forecast the number of arrests made in New York City on a daily basis, using a SARIMAX model. The specifics of the methods we plan to use will be discussed in the next section.

1 Introduction

The overarching goal of time series analysis is to provide a reasonably accurate prediction of future values a measured variable might take, based on its past values. This is particularly useful in a variety of applications, especially when it comes to planning and resource allocation. For instance, a city might wish to allocate police resources efficiently and effectively; that is, preventing waste and over-staffing while ensuring

2 Methods

2.1 Data preparation

The raw dataset contained information on individual arrests made in New York City, including the date of arrest. We grouped this data by date using the pandas library [4] in Python, resulting in a time series of daily arrest counts from January 1, 2012 to December 31, 2017. The final time series contained a total of 2192 data points. We also added a few exogenous variables to the dataset, which we believed would help capture the patterns underlying the vari-

ability in the data. Specifically, we added binary indicators for the outlier days with the lowest arrest counts, as these tend to fall on holidays, or during strikes or other events that disrupt normal city functioning. These crashes, as well as the time series itself, are shown in Figure 1. We will train models both with and without these exogenous variables, to see how much of an impact they have on the model’s performance.

Because the time series exhibited a high degree of variability, we applied a log transformation to the data in an attempt to stabilize the variance. We trained models on both the original and log-transformed data, and compared their performance. The log transformed data is shown in Figure 2.

2.2 SARIMAX

As mentioned in the introduction, time series analysis is a branch of statistics that deals with predicting the future values of a variable based on its past values. The variable of interest in our case is the daily number of arrests made in New York City. We treat this series as a realization of a stochastic process, that is, a collection of random variables indexed by time. One of the most widely used models for time series forecasting is the Autoregressive Integrated Moving Average (ARIMA) model, which captures the temporal dependencies in the data through a combination of autoregressive (AR) and moving average (MA) components, along with differencing (I) to achieve stationarity. While this model is effective for many time series, it may not be sufficient for data exhibiting seasonal patterns, as there may be multiple different differencing operations required to achieve stationarity. Moreover, ARIMA cannot directly incorporate the influence of external factors that may affect the distribution of the variable of interest.

To address this, we use the Seasonal Autoregressive Integrated Moving Average with Exogenous regressors (SARIMAX) model, which extends the ARIMA model to account for seasonal patterns in the data. The SARIMAX model includes additional seasonal AR and MA components, as well as the ability to incorporate exogenous variables that may play a role in influencing the number of arrests made on any given day. These are particularly useful in our case, as we observe a clear weekly seasonality in our data: The autocorrelation function (ACF) shows a repeating peak every 7 days, indicating that the number of arrests made on a given day is correlated with the number of arrests made 7 days prior (See Figure 3). Moreover, the fourier transform of the time series shows a clear peak

at a frequency corresponding to a period of 7 days (See Figure 4).

In total, we will be training four different SARIMAX models on the dataset:

- Model 1: Original time series without exogenous variables. (Called OG-NOEXO)
- Model 2: Original time series with exogenous variables. (Called OG-EXO)
- Model 3: Log-transformed time series without exogenous variables. (Called LOG-NOEXO)
- Model 4: Log-transformed time series with exogenous variables. (Called LOG-EXO)

We use the pmdarima library [2] in Python to fit SARIMAX models to the data. This library provides an ‘auto-arma’ function that automatically selects the optimal hyperparameters for the SARIMAX model based on the Akaike Information Criterion (AIC). We use this function to determine the best values for the non-seasonal (p, d, q) and seasonal (P, D, Q, s) orders of the model (that is, all model parameters other than ‘s’). The value of the seasonal period ‘s’ is set to 7, to capture the weekly seasonality observed in the data.

We will evaluate the performance of each model using a train-test split, where we train the model on 90% of the data and test it on the remaining 10%. We then compute the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), the Mean Absolute Percentage Error (MAPE) and R-squared (R^2) values of the model’s predictions on the test set, to compare the accuracy of the different models. To test the validity of the SARIMAX model assumptions, we also perform the Ljung-Box test on the residuals of each model, to check for any remaining autocorrelation. We also tested the normality of the residuals.

3 Results

Table 1 shows that introducing log transformation plus exogenous indicators (LOG-EXO) yields the lowest MAE (0.10) and RMSE (0.15), highest R^2 (0.6554) and still passes the Ljung-Box check ($p=0.43$), so it best captures the test-set dynamics. OG models on the raw scale have substantially larger errors (MAE $\approx$ 80 - 82, RMSE $\approx$ 110 - 123) despite similar Ljung-Box p-values, indicating residual autocorrelation is under control, but scale makes them less predictive. LOG-NOEXO performs worse than LOG-EXO

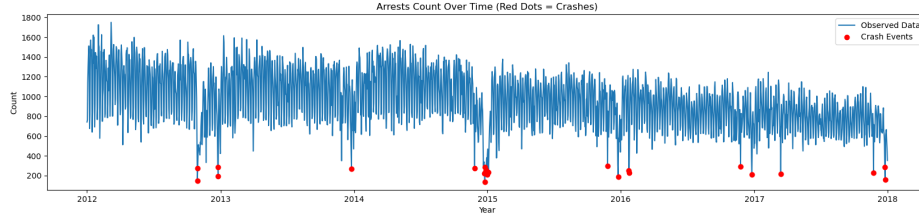


Figure 1: Daily Arrests Time Series with Outlier Days Marked in Red

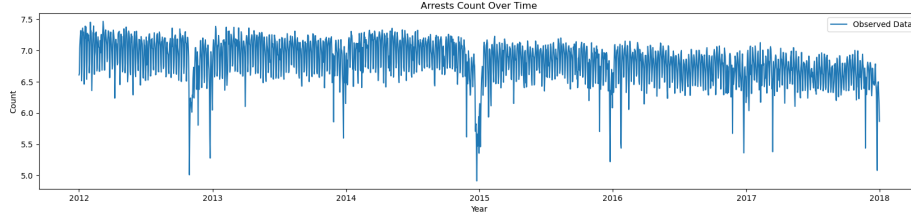


Figure 2: Log-Transformed Daily Arrests Time Series

across all metrics, confirming that the extra regressors improve the log-scale forecasts. Moreover, the Ljung-Box p-value for LOG-NOEXO is 0.05, suggesting some remaining autocorrelation in residuals, indicating the model may not fully capture all patterns without exogenous inputs. All other models have p-values above 0.4, meaning that we fail to reject the null hypothesis of no autocorrelation in residuals at the 5% significance level. Overall, LOG-EXO emerges as the best-performing model, demonstrating the benefits of both log transformation and inclusion of exogenous variables in accurately forecasting daily arrests. However, due to the high autocorrelations observed in the residuals of the LOG-EXO model as seen in Figure 8, we choose to use the OG-EXO model for further analysis. Despite its slightly worse performance metrics compared to LOG-EXO, the OG-EXO model exhibits lower autocorrelations in the residuals, indicating a better fit to the underlying data patterns.

We tested for normality of the residuals by plotting Q-Q plots, a histogram of the residuals, and the autocorrelation function (ACF) of the residuals. These plots can be found in Figures 5, 6, 7, 8 for the OG-NOEXO model, OG-EXO model, LOG-NOEXO model, and LOG-EXO model respectively.

We have also plotted the predictions of each model on the test set, along with the actual values, in Figures 9, 10, 11 and 12 for the OG-NOEXO model, OG-EXO model, LOG-NOEXO model, and LOG-EXO model respectively.

4 Discussion

The primary objective of this study was to develop a reliable time series model to forecast daily criminal arrests in New York City, thereby facilitating more efficient police resource allocation. We evaluated four SARIMAX model variations, distinguishing between original vs. log-transformed data and the inclusion vs. exclusion of exogenous variables.

The comparison between "EXO" and "NOEXO" models confirms the necessity of accounting for external disruptions. The inclusion of binary indicators for outlier days (representing holidays, strikes, or other significant city events) consistently improved model performance. For instance, the LOG-NOEXO model performed worse across all metrics compared to LOG-EXO and failed the Ljung-Box test ($p=0.05$), indicating that without these external regressors, the model could not account for significant variance in the data. This implies that arrest rates cannot be explained solely by a fixed non-conditional distribution, but are heavily influenced by social and civic events.

To address the high degree of variability in the daily arrest data, we initially applied a log transformation to stabilize the variance. The resulting model (LOG-EXO) demonstrated superior statistical performance, achieving the lowest Mean Absolute Error (0.10) and Root Mean Squared Error (0.15), as well as the highest R2 value of 0.6554. However, we ultimately decided against using this transformed model for our final analysis. Despite passing the Ljung-Box test ($p=0.43$), a visual inspection of the residuals (Figure 8) revealed significantly high autocorrelations that were not present in the model

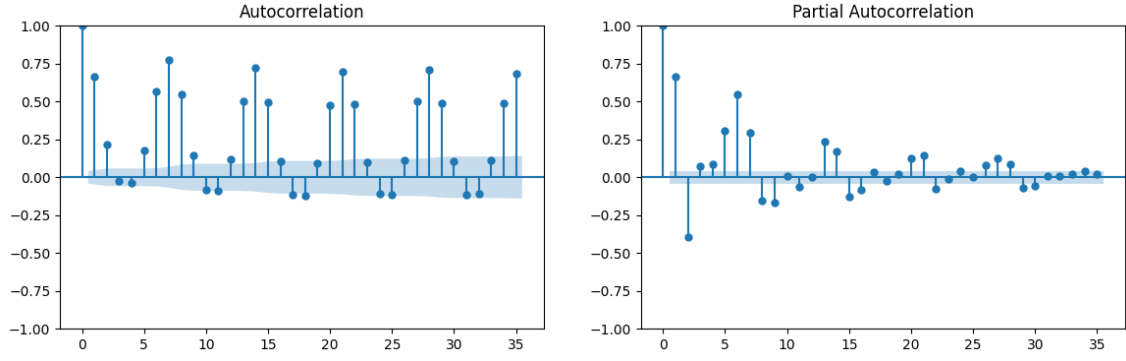


Figure 3:

Model	MAE	RMSE	MAPE (%)	R^2	Ljung-Box p-value
OG-NOEXO	82.16	122.61	14.19%	0.4403	0.64
OG-EXO	78.77	109.09	12.08%	0.5569	0.64
LOG-NOEXO	0.11	0.2	1.76%	0.3748	0.05
LOG-EXO	0.1	0.15	1.59%	0.6554	0.43

Table 1: Model Performance Metrics on Test Set

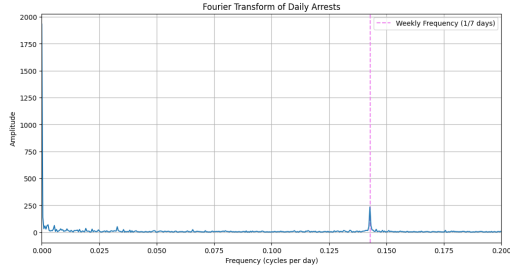


Figure 4: Fourier Transform of Daily Arrests Time Series

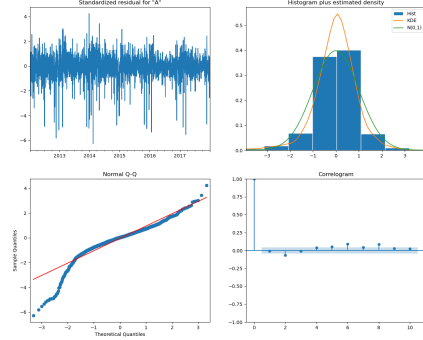


Figure 5: Residual Analysis for the OG-NOEXO Model

trained on the original scale. This discrepancy suggests that while the log-transformed model minimized numerical error, the Original-Scale (OG-EXO) model provided a better structural fit to the underlying data patterns, as evidenced by its lower residual autocorrelation.

Even a cursory glance at these plots shows that there are significant autocorrelations present in the residuals of all models. The Q-Q plots also show significant deviations from the straight line, indicating that the residuals are not normally distributed. This suggests that while the SARIMAX models are able to capture some of the patterns in the data, there are still underlying structures that are not being accounted for. Further investigation and potentially more sophisticated modeling techniques may be required to fully capture the dynamics of the time series.

Conclusions

We evaluated four SARIMAX model variations, finding that incorporating exogenous indicators for outlier days substantially improved predictive accuracy. While log-transforming the data reduced error metrics, the original scale model provided a better structural fit, as evidenced by lower residual autocorrelation. The OG-EXO model achieved a Mean Absolute Error of 78.77, and an R^2 of 0.5569 on the test set, indicating moderate predictive capability. This study demonstrates that time series forecasting can be a powerful tool for law enforcement agencies seeking to optimize their operations. By aggregating NYPD arrest data from 2012 to 2017, we established that daily arrest counts exhibit strong weekly seasonality and are significantly

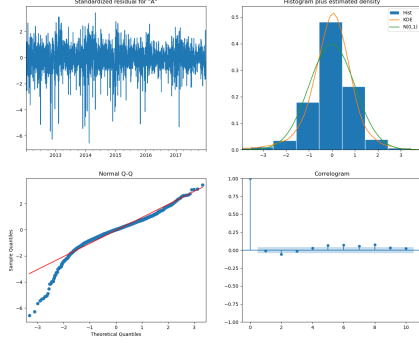


Figure 6: Residual Analysis for the OG-EXO Model

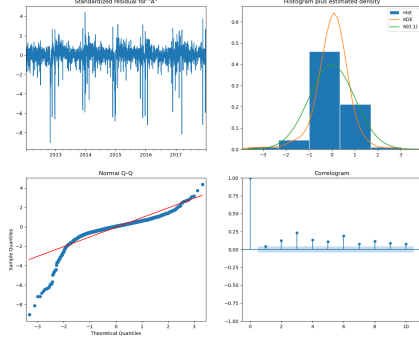


Figure 7: Residual Analysis for the LOG-NOEXO Model

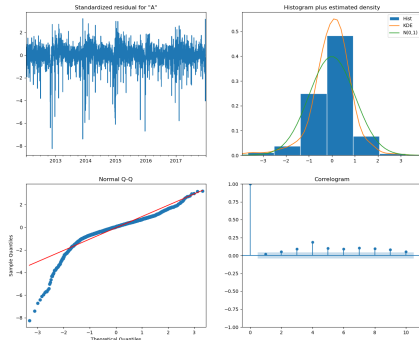


Figure 8: Residual Analysis for the LOG-EXO Model

impacted by specific calendar events.

As seen above, while SARIMAX models can capture some patterns in the data and have some predictive power, there are still underlying structures not accounted for, as evidenced by the significant autocorrelations in the residuals. Future work could explore more sophisticated modeling techniques, such as non-linear models and approaches, to better capture the complex dynamics of the dataset.

References

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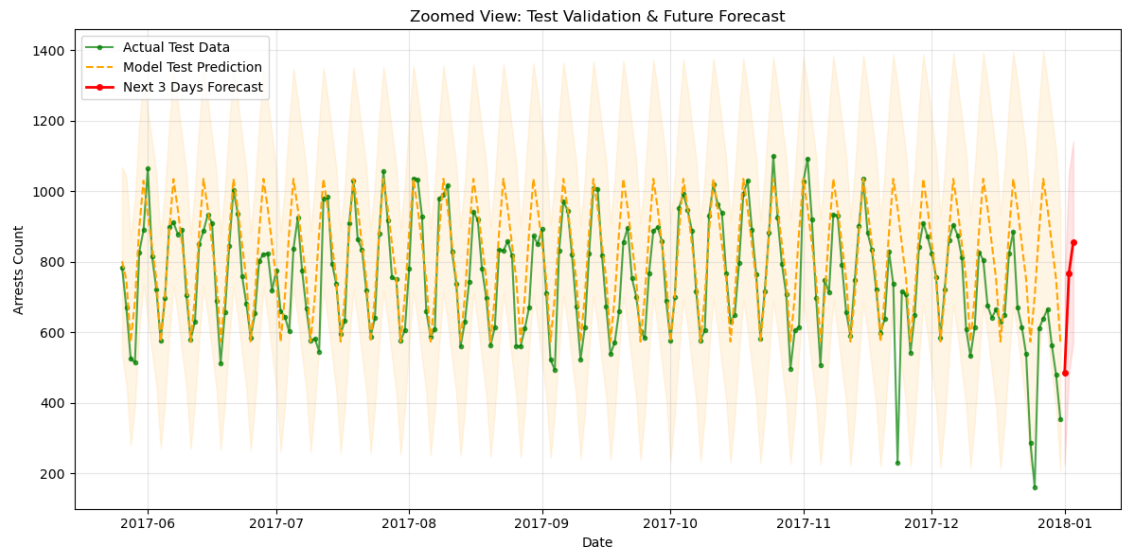


Figure 9: OG-NOEXO Model Predictions vs Actuals on Test Set

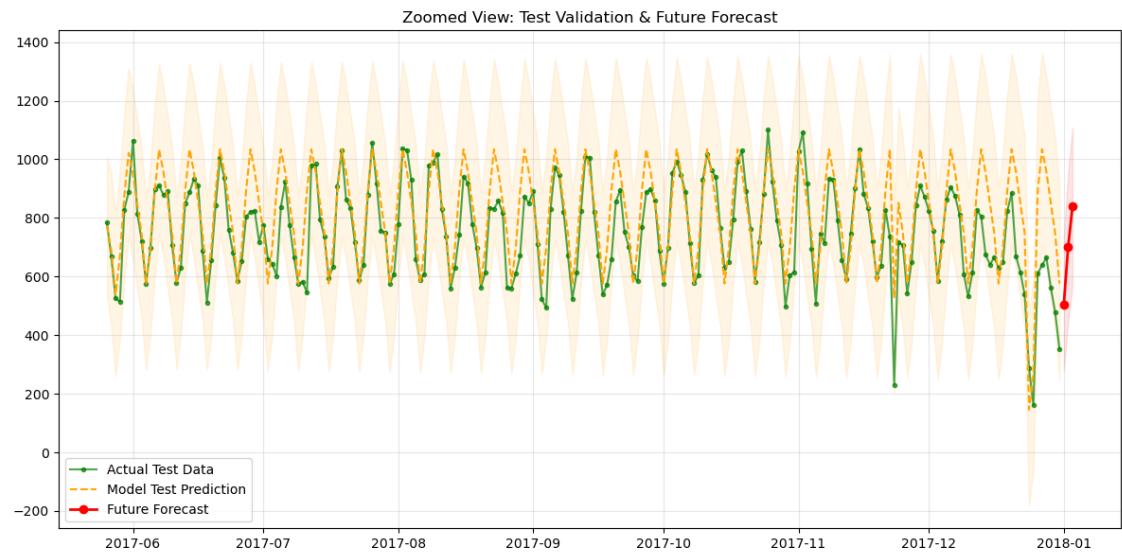


Figure 10: OG-EXO Model Predictions vs Actuals on Test Set

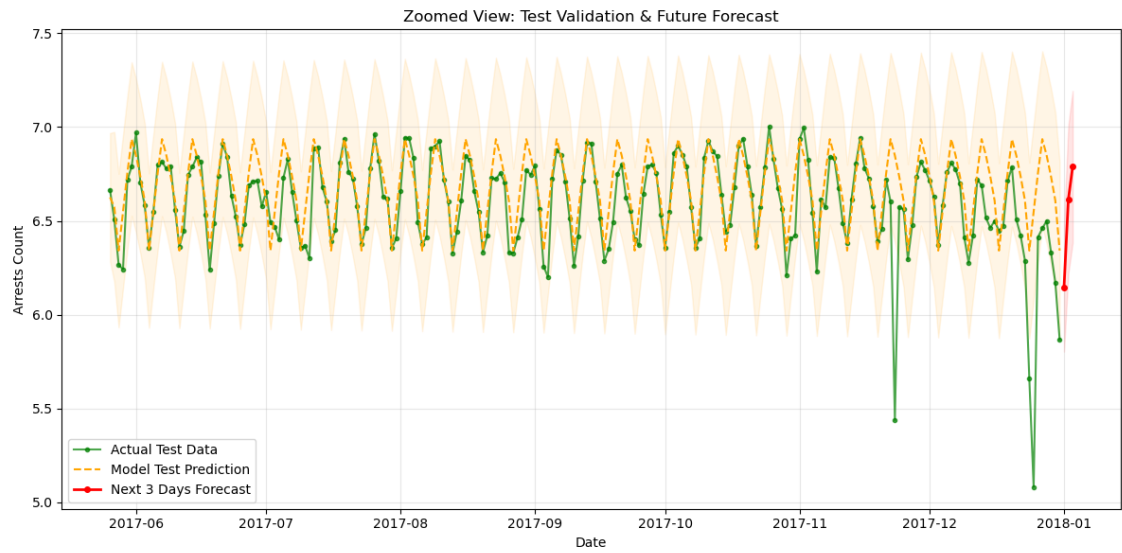


Figure 11: LOG-NOEXO Model Predictions vs Actuals on Test Set

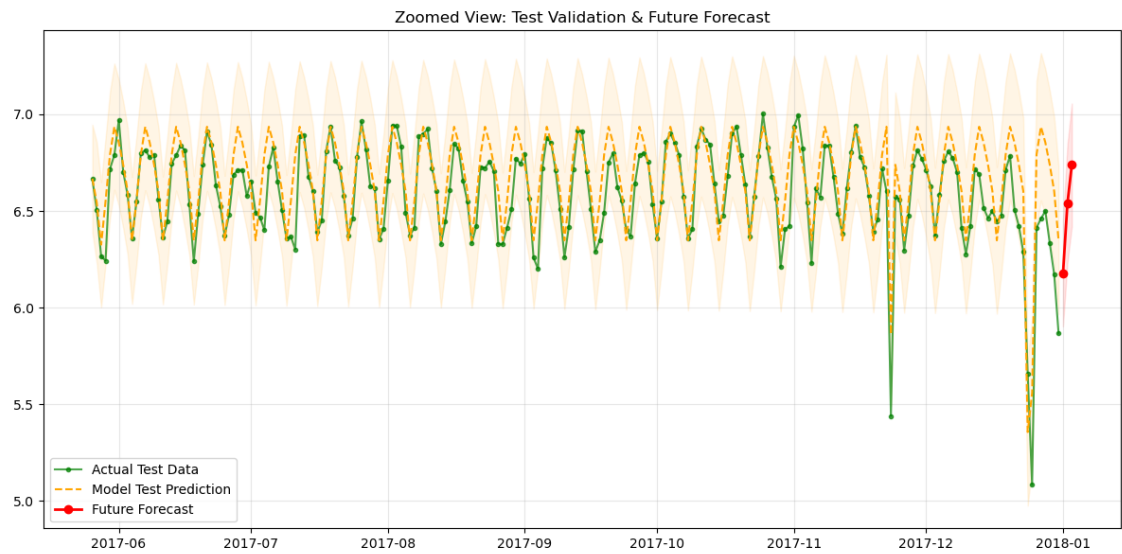


Figure 12: LOG-EXO Model Predictions vs Actuals on Test Set